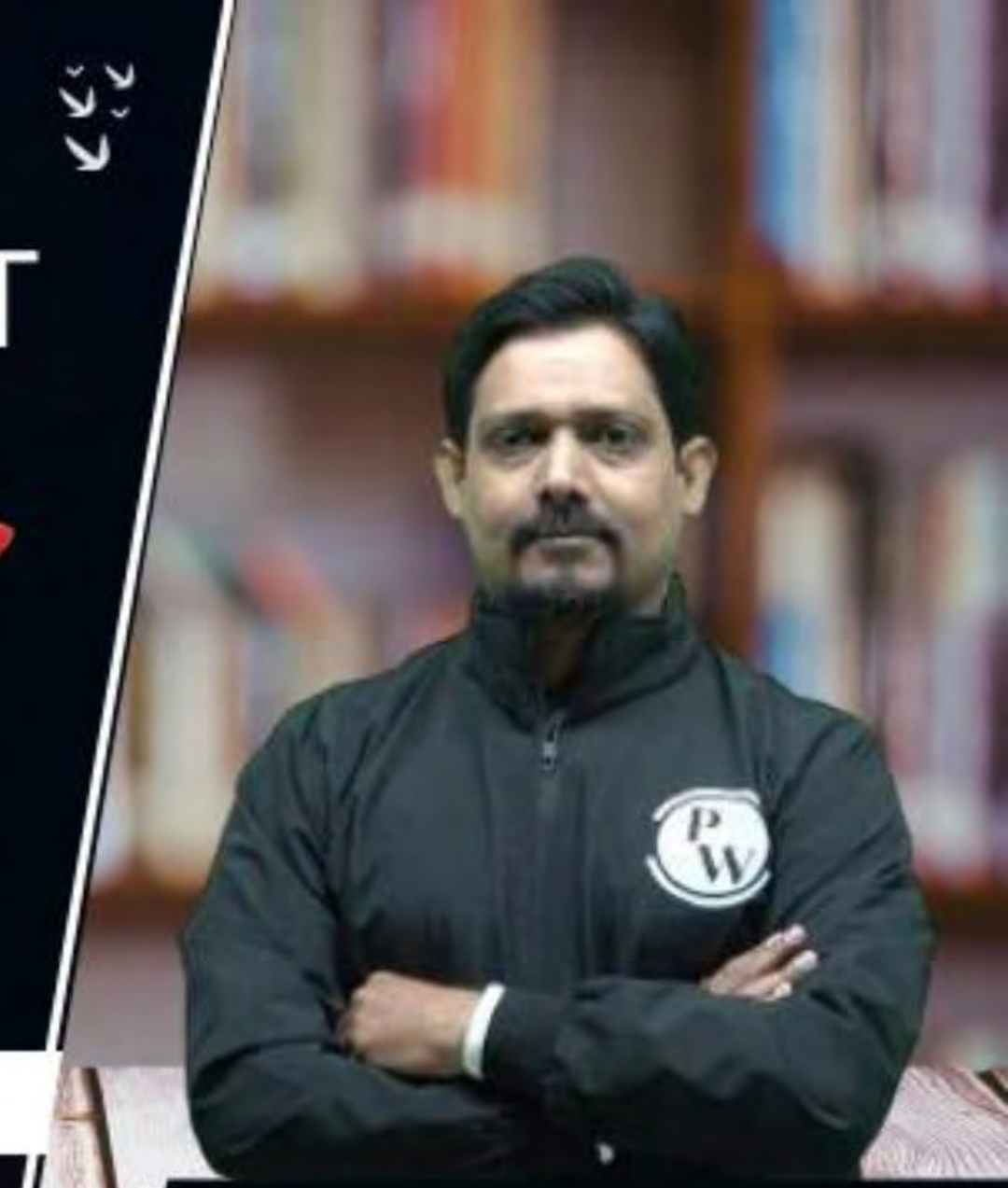
Computer Science & IT



Data Structure

Tree

Lecture No. 08



By- Abhishek Sir

Recap of Previous Lecture



Topic — AVL tree Rotation

Topic — RR LL

Topic — LR , RL

Topic — Minimum No. of Node in height h

$$N(h) = N(h-1) + N(h-2) + 1$$

Topic

Slide

Topics to be Covered



Topic	priority queue
Topic	Heap,
Topic	Max heap
Topic	Min Heap
Topic	Insert, Delete, Adjust

Slide



Topic : Tree

Max heap min heap
Heap Insertion
Heap Deletion
Adjust
Build heap
Counting

Slide



Topic : Heap

priority queue : In priority queue we can insert element in any order but deletion will done based on priority.

one way to Implement priority queue is using binary heap



Topic : Heap

An element with high priority is dequeued before an element with low priority.



Topic : Heap

Max heap and Min Heap Data Structure.

Complete binary tree

Application 1. Greedy Method Optimization

job sequencing

Shortest path (Min Heap)

Spanning tree (prims)



Topic : Heap

Applications of Priority Queue:

CPU Scheduling

Graph algorithms like Dijkstra's shortest path algorithm, Prim's Minimum Spanning Tree, etc

All queue applications where priority is involved.



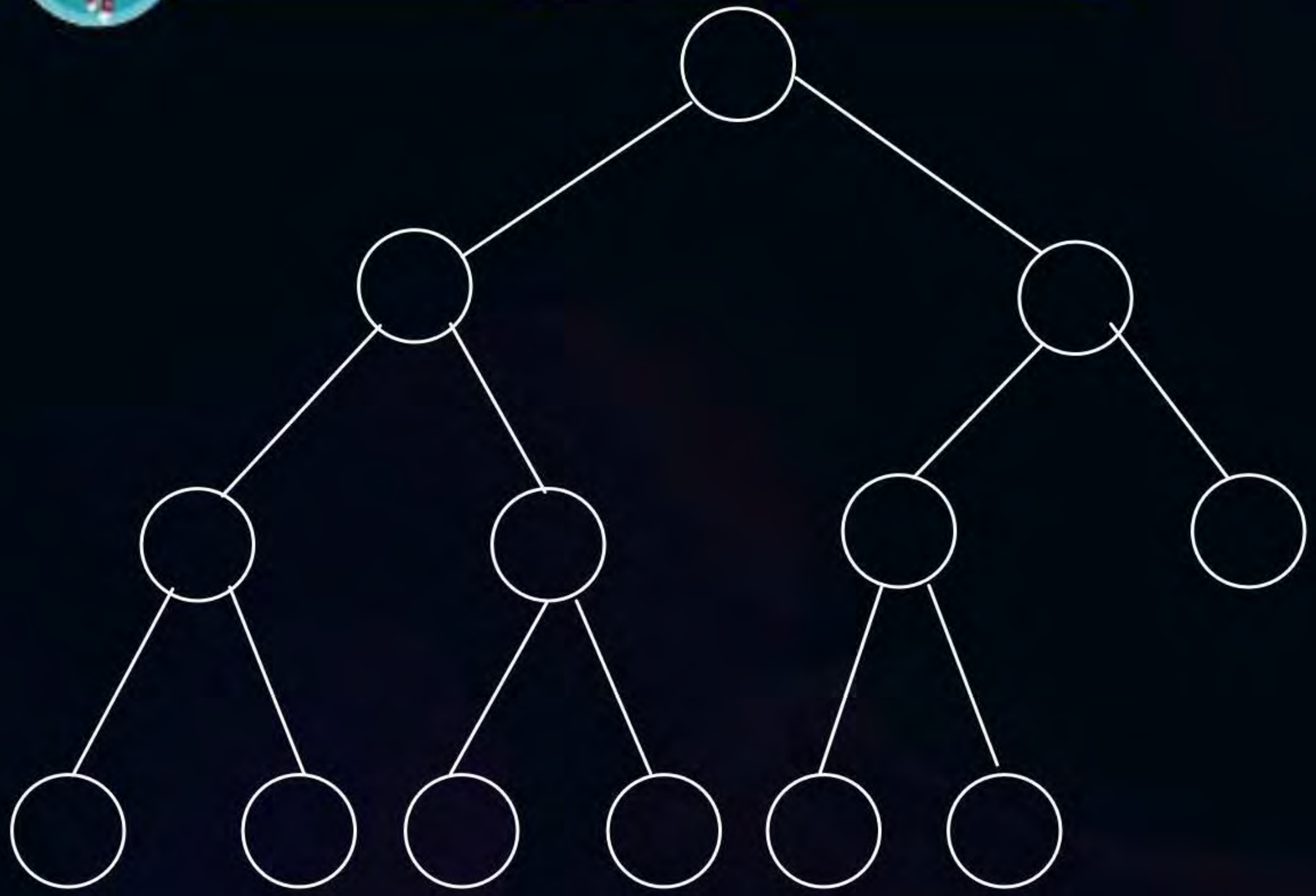
Topic : Heap

Heap is Complete binary tree

CBT is a tree in which every level is full except possibly the last. In Last level elements are arranged strictly from left to right.



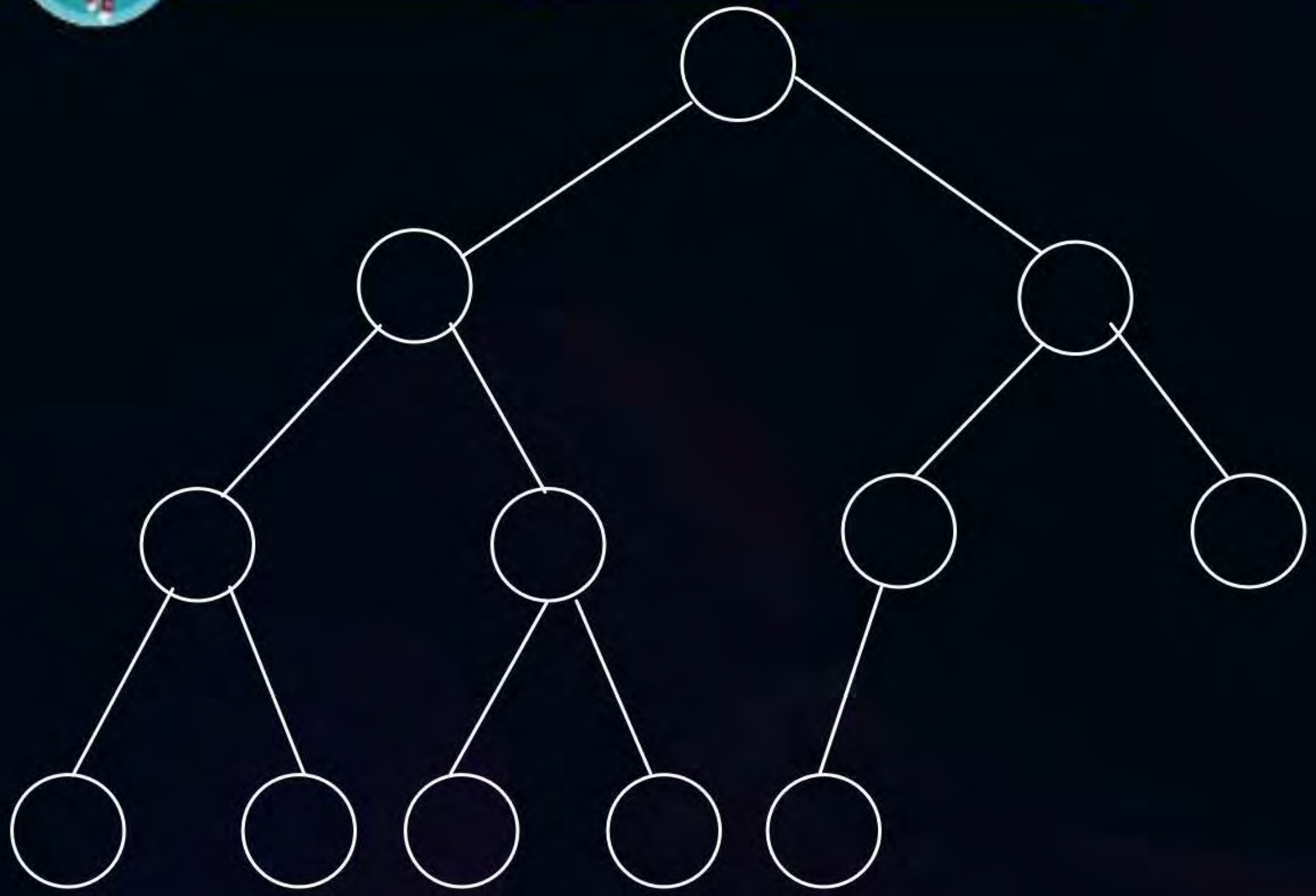
Topic : Heap



yes the structur CBT



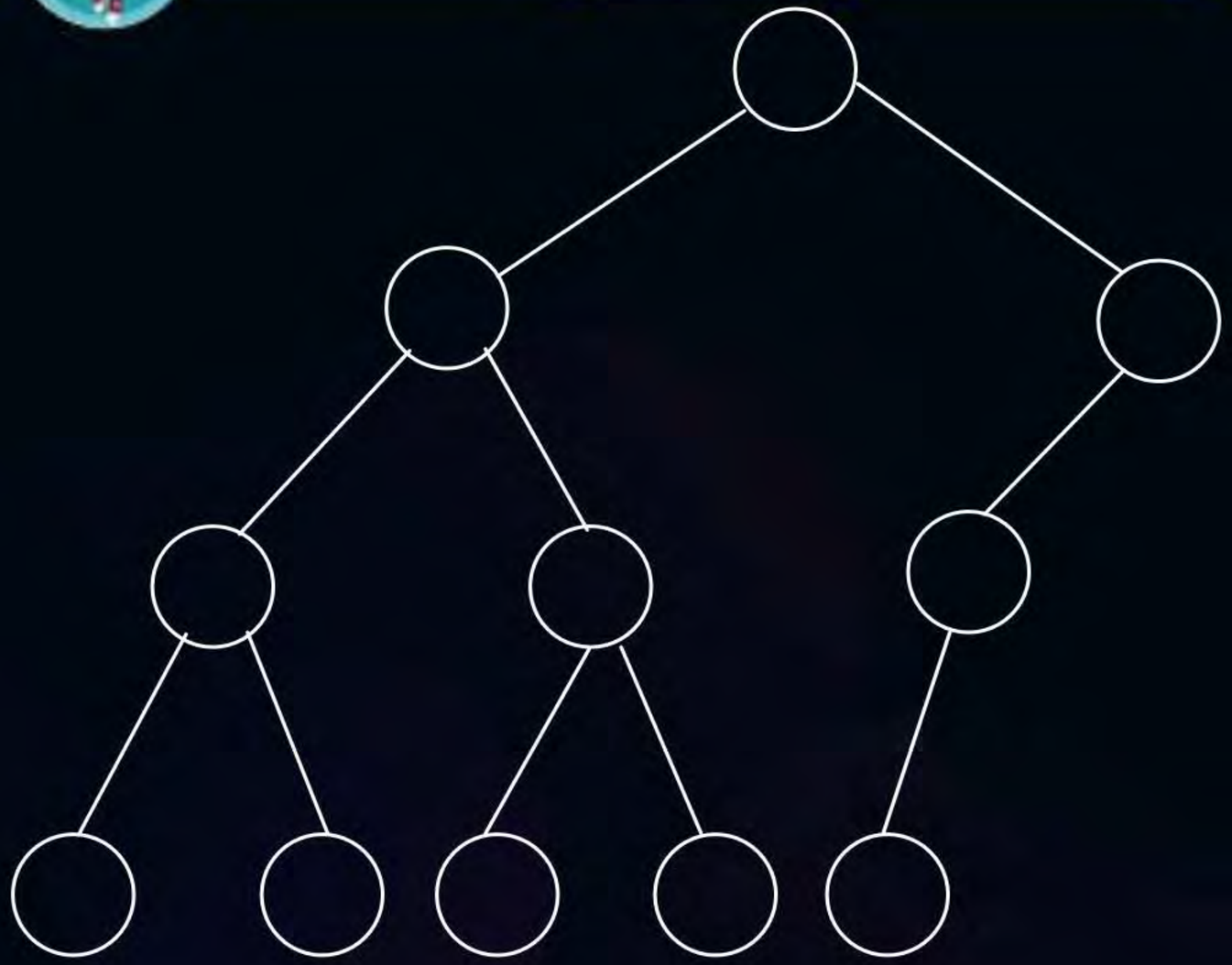
Topic : Heap



yes the structur CBT



Topic : Heap



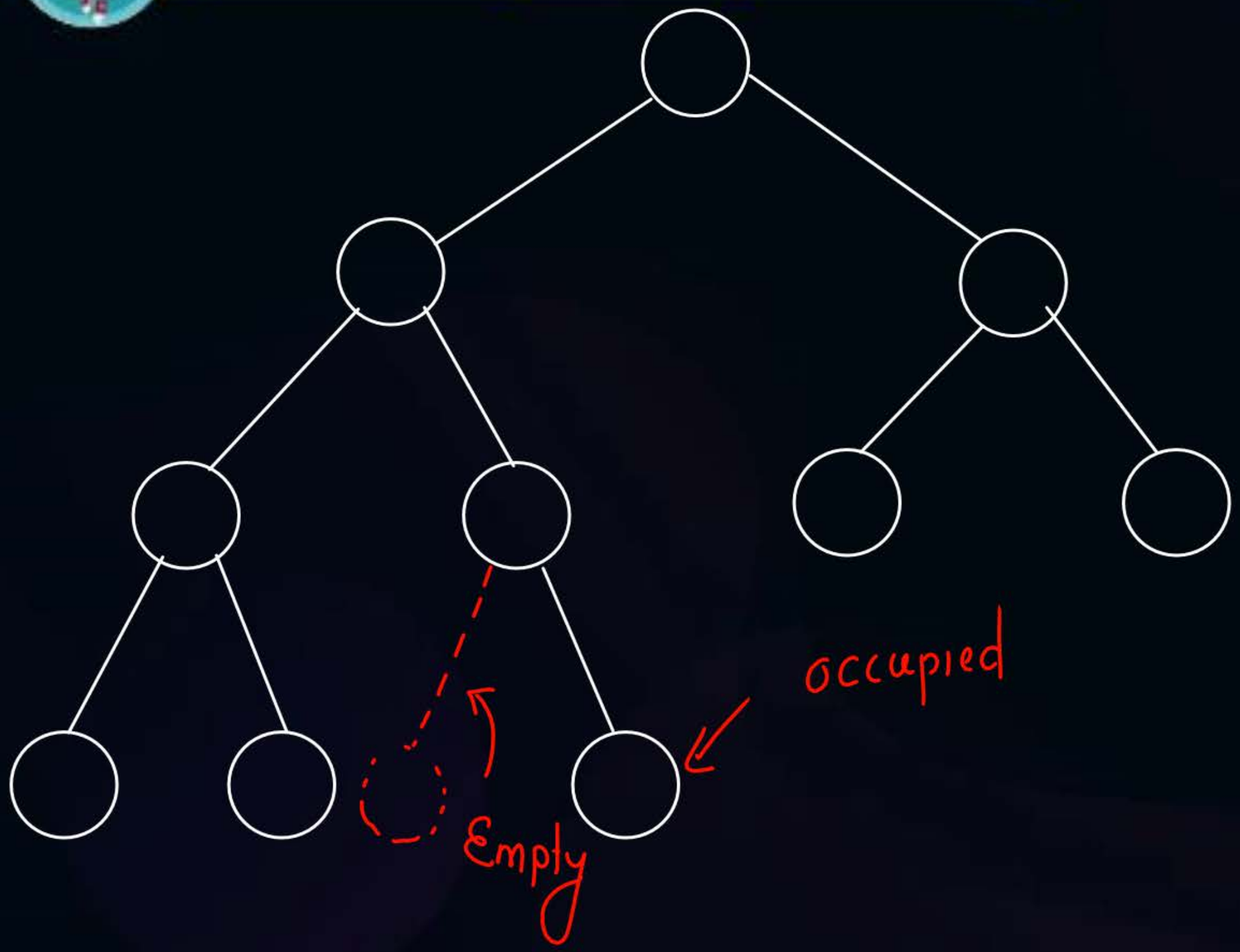
No

The level before Last

Level is NOT Full



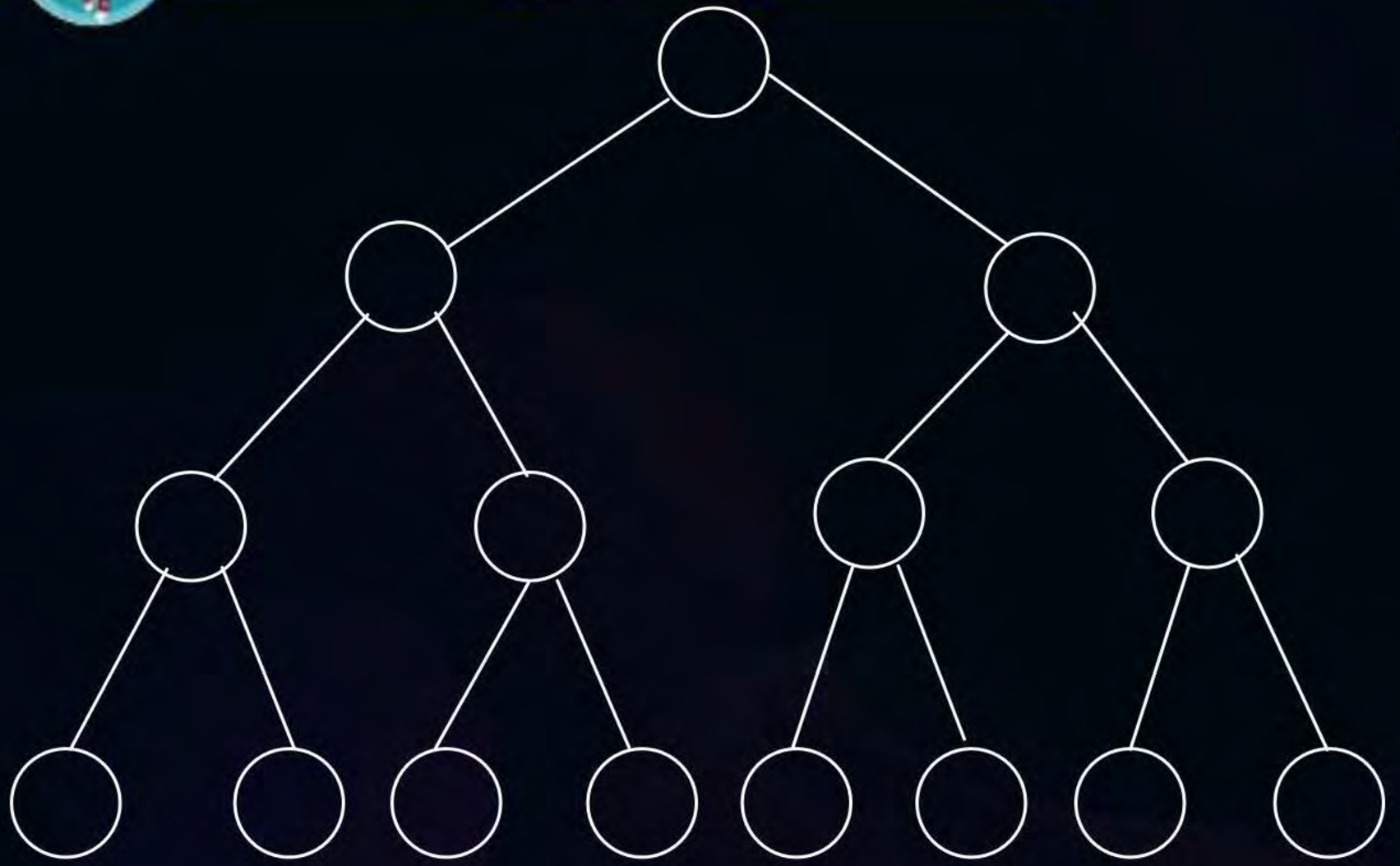
Topic : Heap



Not a CBT structure



Topic : Heap



yes the structur CBT



Question

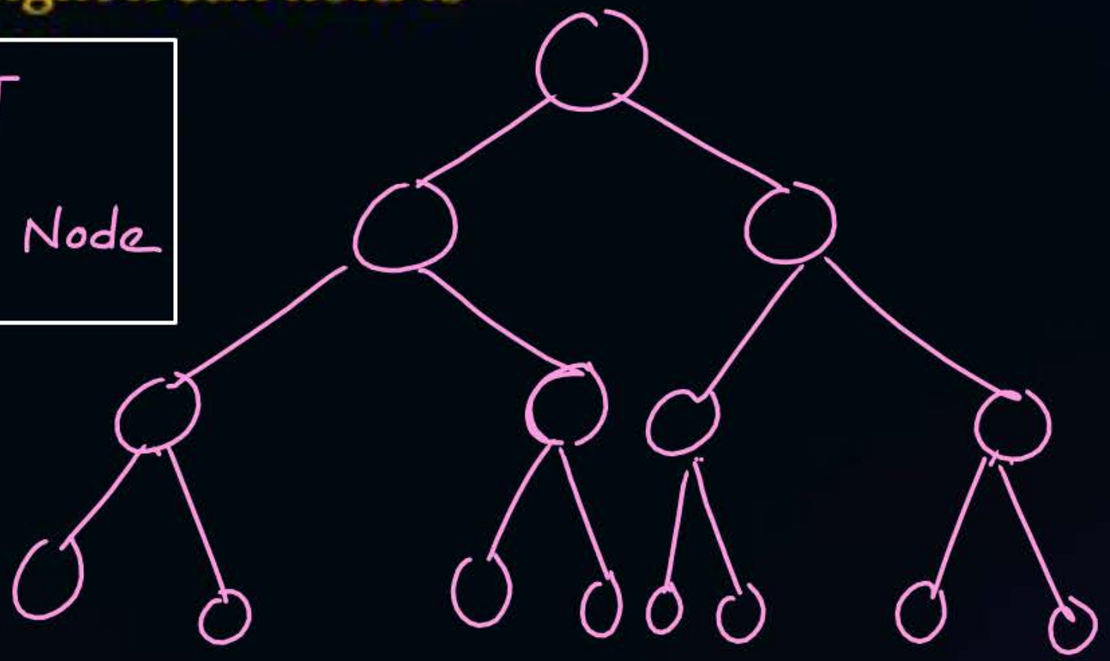
#Q. The number of nodes a heap of height k can hold is

[A]

Max care - FBT

Min. heigh h 1 Node

$2^{3+1} - 1 = 15$



(A) 2^k to $2^{k+1} - 1$

(B) 2^{k+1} to $2^{k+1} - 1$

(C) 2^{k-1} to $2^{k+1} - 1$

(D) $2^k - 1$ to $2^{k+1} - 1$



Question

$2^{3+1}-1-1+1$

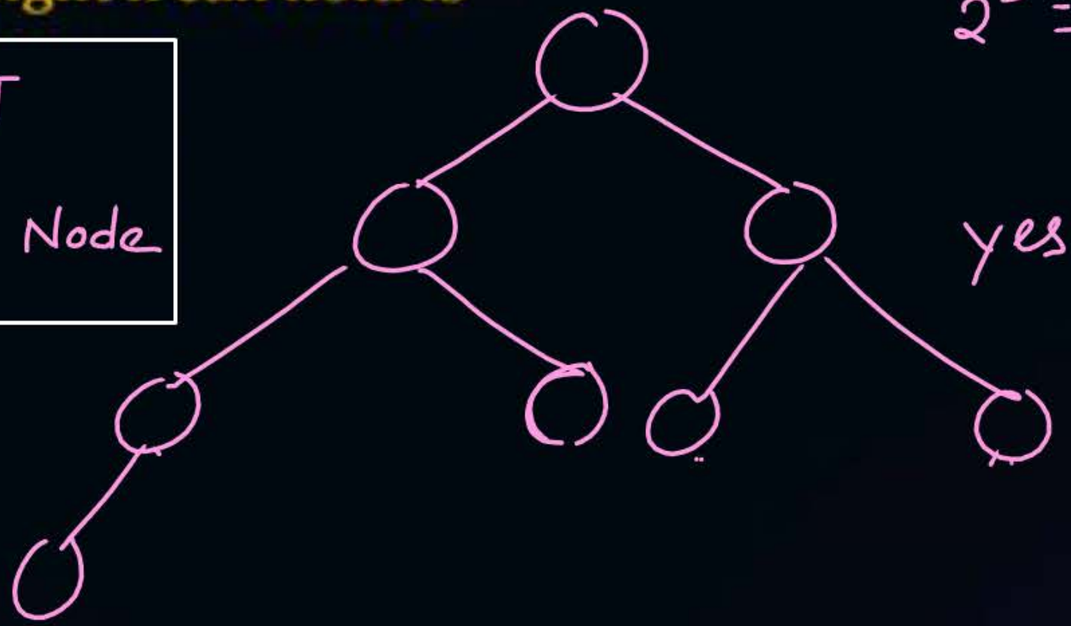
$2^3 = 8$

#Q. The number of nodes a heap of height k can hold is

[A]

Max care - FBT

Min. heigh h 1 Node



yes

(A) 2^k to $2^{k+1}-1$

(B) 2^{k+1} to $2^{k+1}-1$

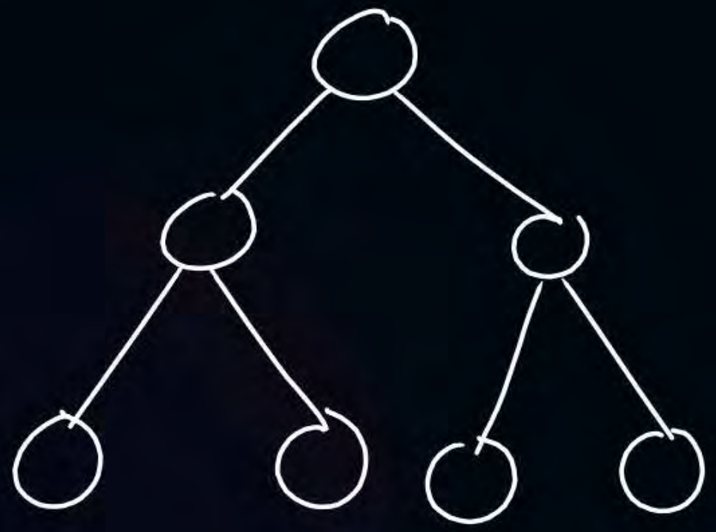
(C) 2^{k-1} to $2^{k+1}-1$

(D) $2^k - 1$ to $2^{k+1}-1$



Topic : Heap

Why CBT structure



FBT $h = \log_2(n+1) - 1$

$= \log_2 8 - 1 = 3 - 1 = 2$

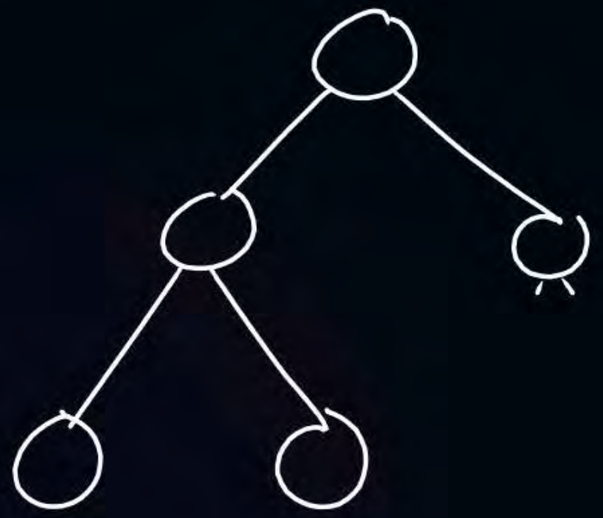
CBT height $= \lfloor \log_2 n \rfloor$

$= \lfloor \log_2 7 \rfloor = 2$



Topic : Heap

Why CBT structure



FBT $h = \log_2(n+1) - 1$

$= \log_2 8 - 1 = 3 - 1 = 2$

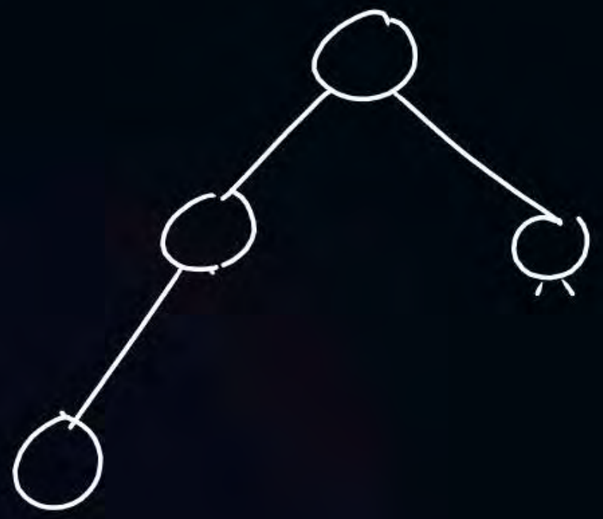
CBT height $= \lfloor \log_2 n \rfloor$

$= \lfloor \log_2 5 \rfloor = 2$



Topic : Heap

Why CBT structure



FBT $h = \log_2(n+1) - 1$

$= \log_2 8 - 1 = 3 - 1 = 2$

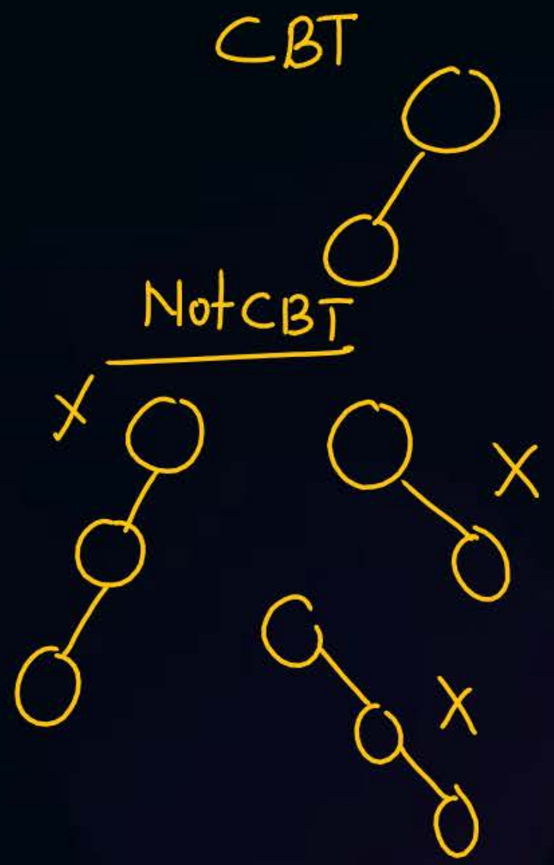
CBT height : $\lfloor \log_2 n \rfloor$

$\lfloor \log_2 4 \rfloor = 2$



Topic : Heap

CBT gives minimum Height with n-nodes



$$h = \lfloor \log_2 n \rfloor$$

$$= \lfloor \log_2 26 \rfloor = 4$$



Topic : Complete Binary Tree

Height of complete Binary tree restricted = $\lfloor \log_2 n \rfloor$



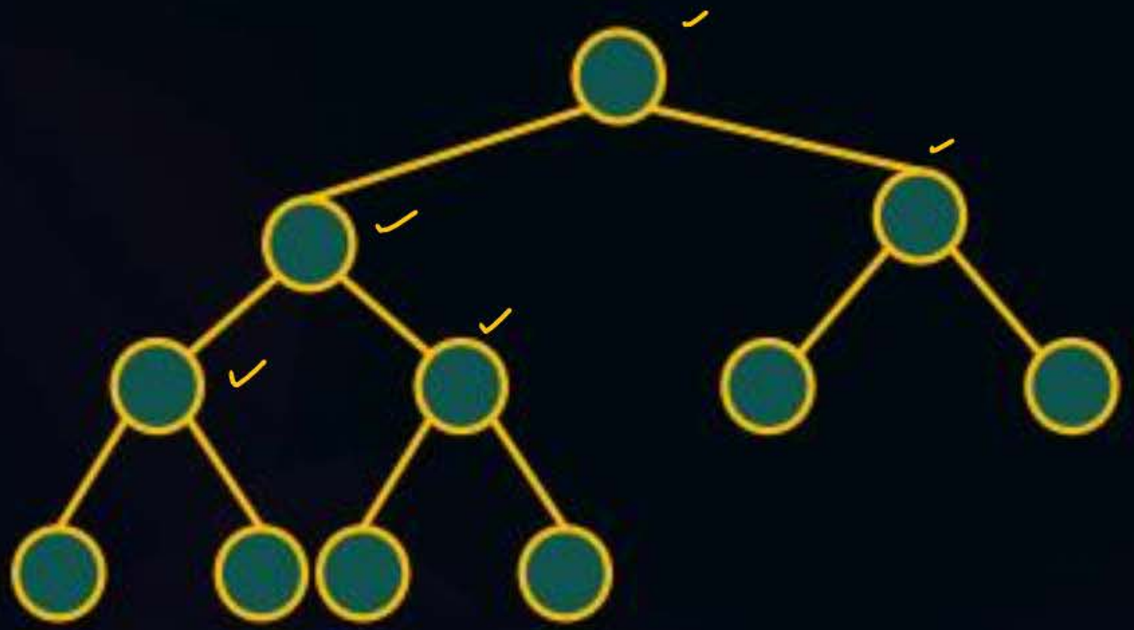
No. of internal Node = $\lfloor \frac{n}{2} \rfloor$

$= \lfloor \frac{8}{2} \rfloor = 4$



Topic : Heap

Number of internal nodes is $= \left\lfloor \frac{n}{2} \right\rfloor$ $= \left\lfloor \frac{11}{2} \right\rfloor = 5$ internal



No. of leaves

$= \left\lceil \frac{11}{2} \right\rceil = 6$

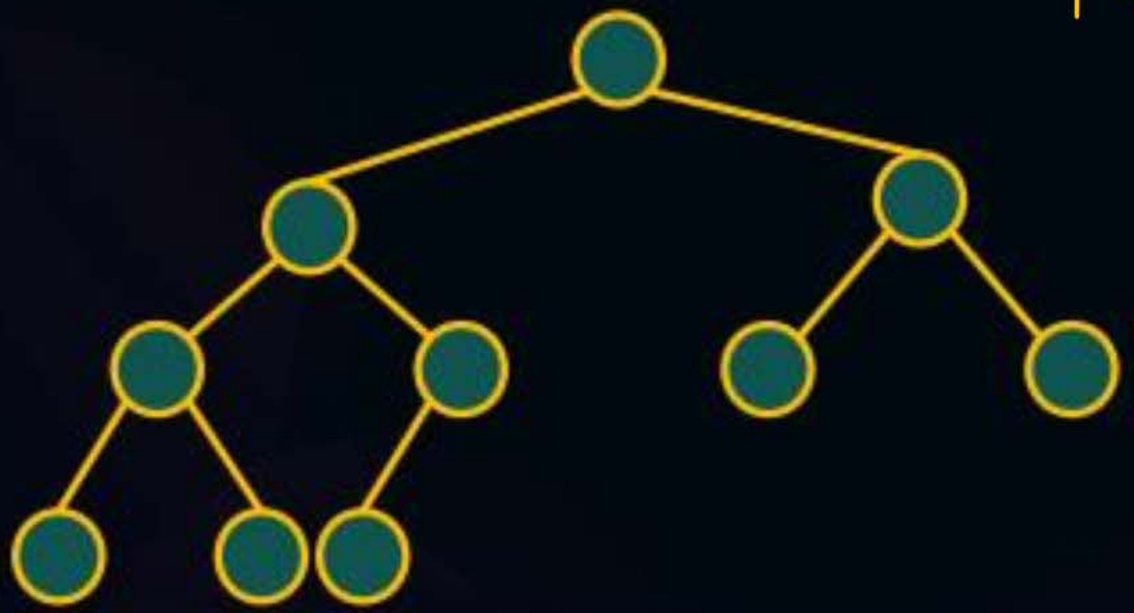


Topic : Heap

ped ka patta upper Hai

Number of leaves is :- $\lceil \frac{n}{2} \rceil$

$\lceil \frac{10}{2} \rceil = 5$





Heap is Not BST

Max Heap : Max heap is CBT in which the key value at parent Node is greater than value at its children

This property true for subtree as well



Topic : Heap

Min Heap :- Min Heap is CBT in which value at parent node is Lesser than value at the children node

This property true for subtree as well.



Topic : Min Heap

Min Heap



Min Heap

Min - Root

Maximum element

will be present

Leaf Node



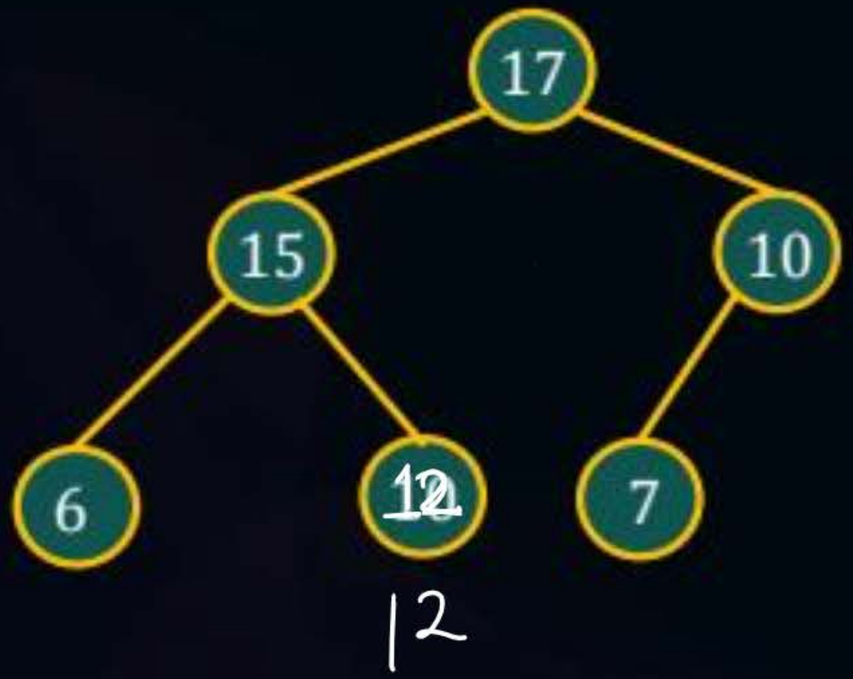
Topic : Max Heap

Max-Heap: In a Max-Heap the key present at the root node must be greatest among the keys present at all of it's children. The same property must be recursively true for all sub-trees in that Binary Tree.



Topic : Heap

Max-Heap



Max heap

Root: Maximum value

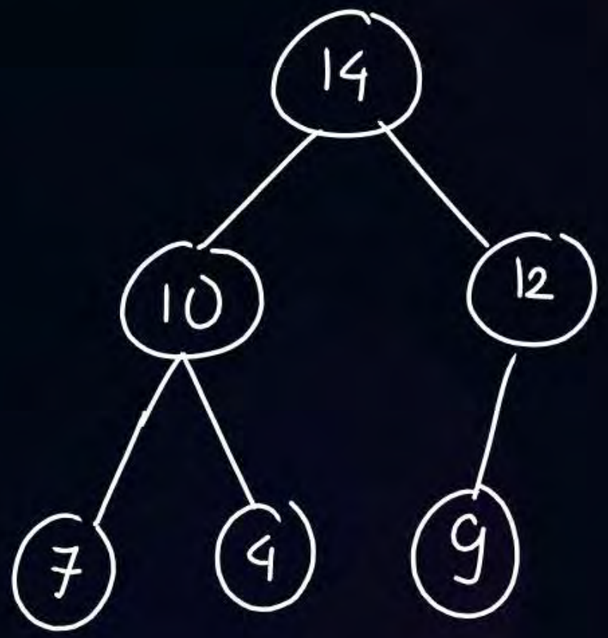
Minimum key

Leaves



Topic : Complete Binary Tree

A complete binary tree is a binary tree in which every level, except possibly the last, is completely filled, and all nodes are as far left as possible.



Heap : Sequential representation

Heap array CBT minimum Space

1	2	3	4	5	6
14	10	12	7	4	9



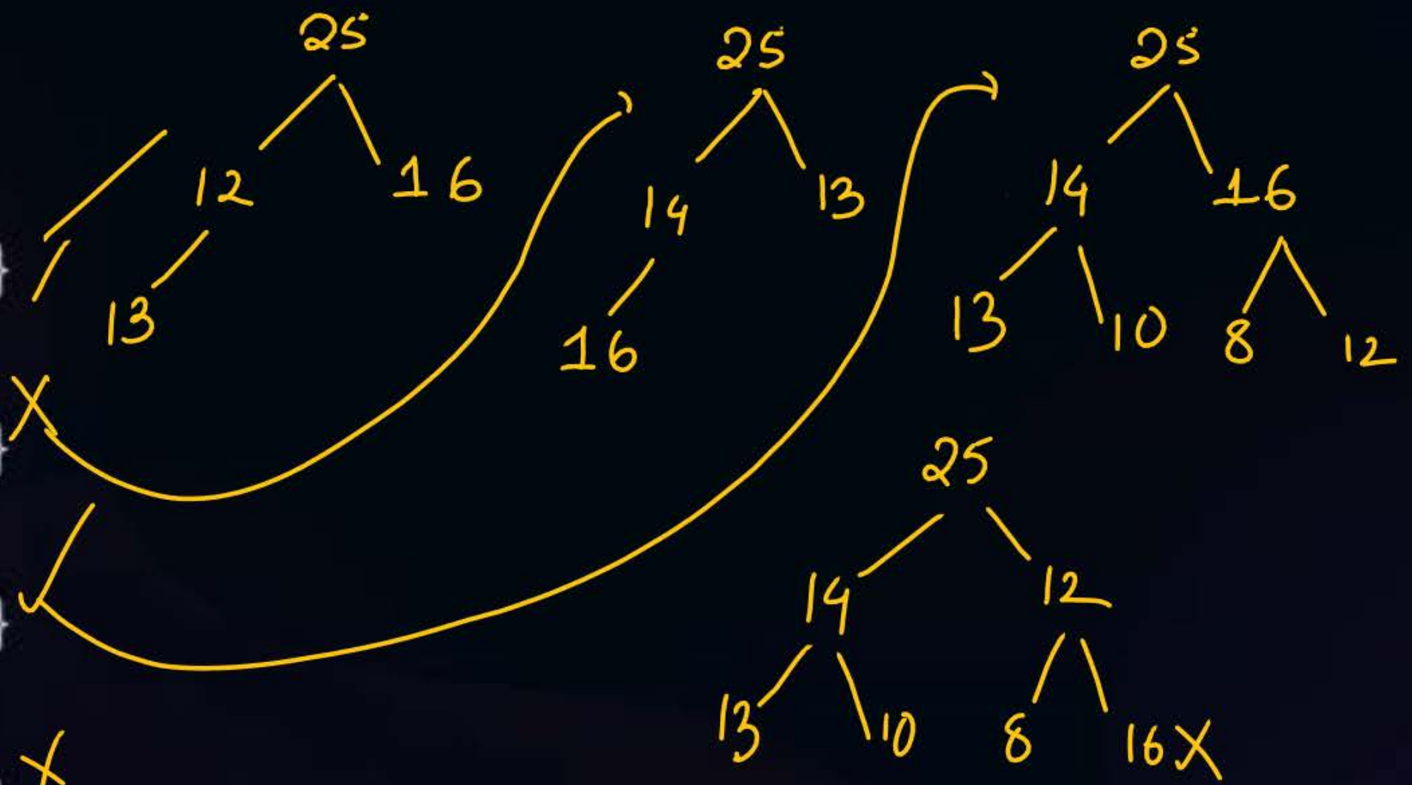
Question

#Q. Consider a binary min-heap containing 169 distinct elements. Number of leaves present in heap is _____

$$\left\lceil \frac{169}{2} \right\rceil = \lceil 84.5 \rceil = 85$$

Question

#Q. Consider a binary max-heap implemented using an array. Which one of the following array represents a binary max-heap?



(A) {25, 12, 16, 13, 10, 8, 14}

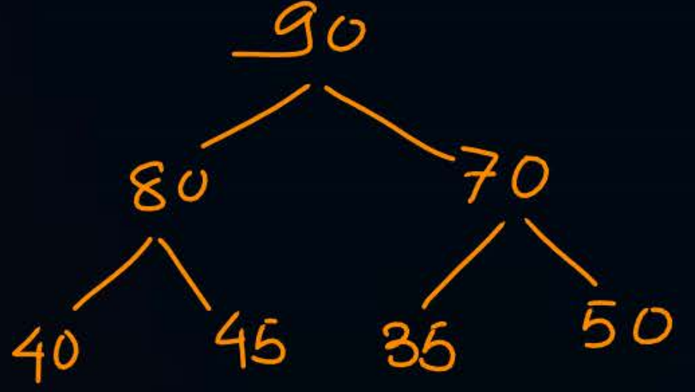
(B) {25, 14, 13, 16, 10, 8, 12}

(C) {25, 14, 16, 13, 10, 8, 12}

(D) {25, 14, 12, 13, 10, 8, 16}



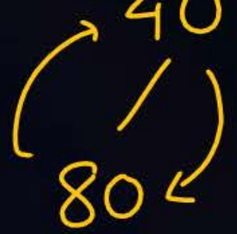
Heap Building by insertion



Forming a max heap from the set
{40,80,35,90,45,50,70}

(I) 40

(II) 80



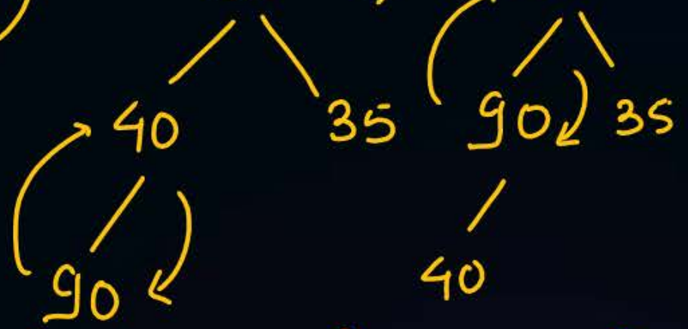
⇒



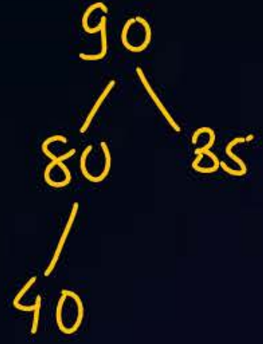
(III)

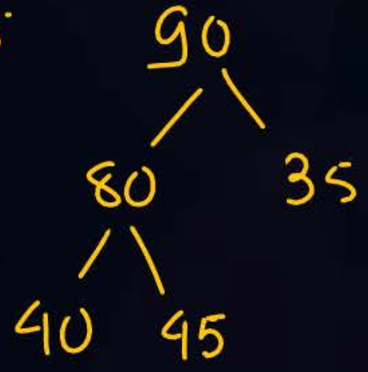


(IV)

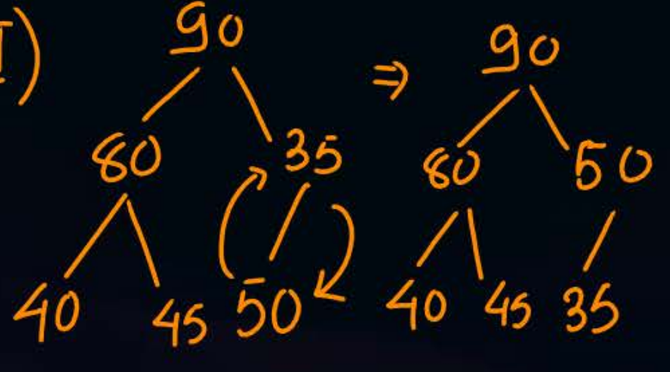


(V) 45

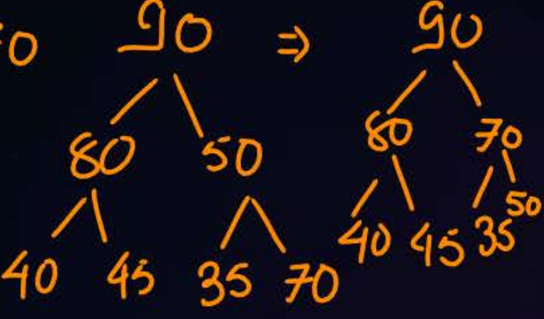




(VI)



(VII) 70





Heap Building by insertion

Forming a max heap from the set
{40,80,35,90,45,50,70}



Newly inserted element
in Last available position for CBT

the we swap it with parent Node

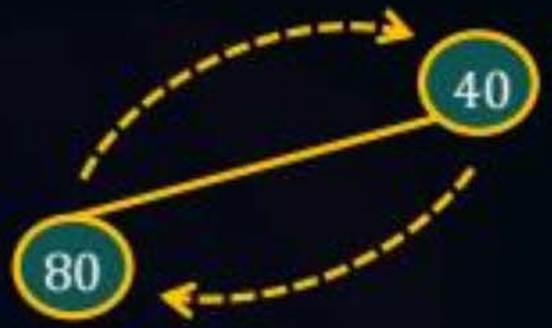
if parent value is lesser (Max heap)
Greater (Min Heap) this

process repeated until correct
position of Newly inserted element found



Topic : Heap

Forming a max heap from the set
{40,80,35,90,45,50,70}





Heap Building by insertion

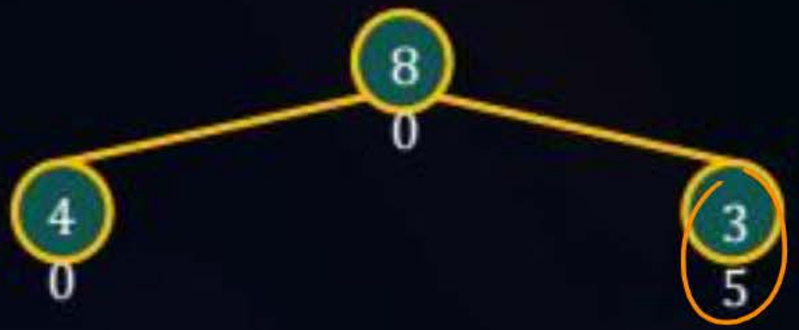
Forming a max heap from the set
{40,80,35,90,45,50,70}





Heap Building by insertion

Forming a max heap from the set
{40,80,35,90,45,50,70}





Heap Building by insertion

Forming a max heap from the set
{40,80,35,90,45,50,70}





Heap Building by insertion

Forming a max heap from the set
{40,80,35,90,45,50,70}





Heap Building by insertion

Forming a max heap from the set
{40,80,35,90,45,50,70}





Heap Building by insertion

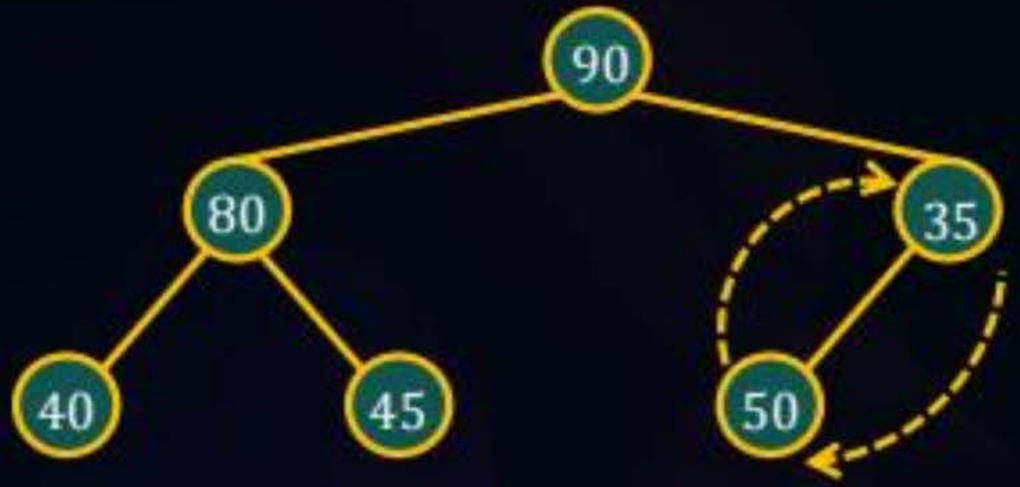
Forming a max heap from the set
{40,80,35,90,45,50,70}





Heap Building by insertion

Forming a max heap from the set
{40,80,35,90,45,50,70}





Heap Building by insertion

Forming a max heap from the set
{40,80,35,90,45,50,70}



Heap Building by insertion

Forming a max heap from the set
{40,80,35,90,45,50,70}



Max swap for newly inserted element is depends upon Height hence Insertion required

$O(\log_2 n)$



Heap Building by insertion

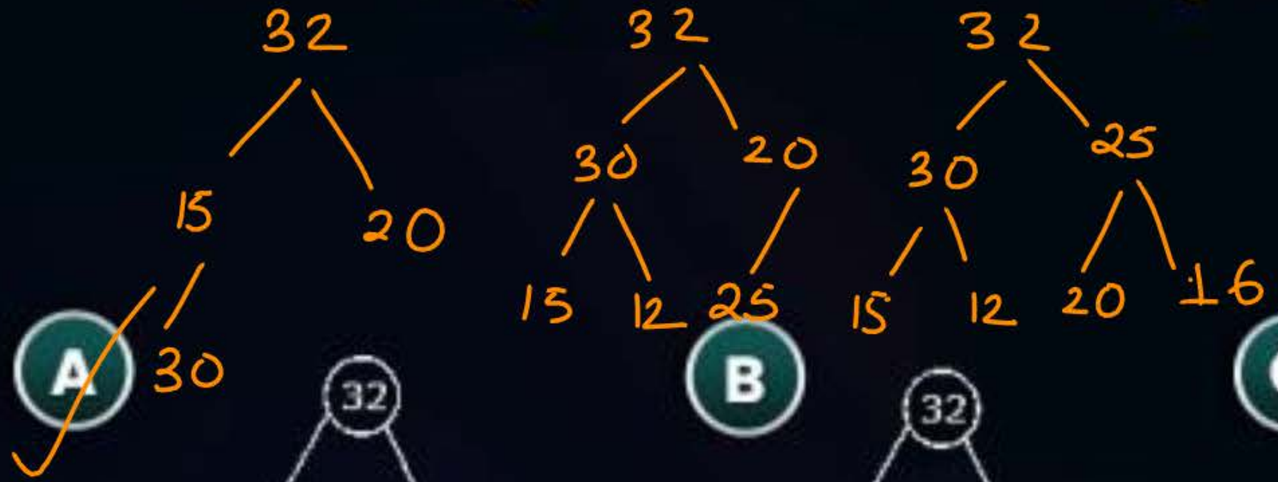
Forming a max heap from the set
{40,80,35,90,45,50,70}





Topic : Heap

#Q. The elements 32, 15, 20, 30, 12, 25, 16, are inserted one by one in the given order into a maxHeap. The resultant maxHeap is



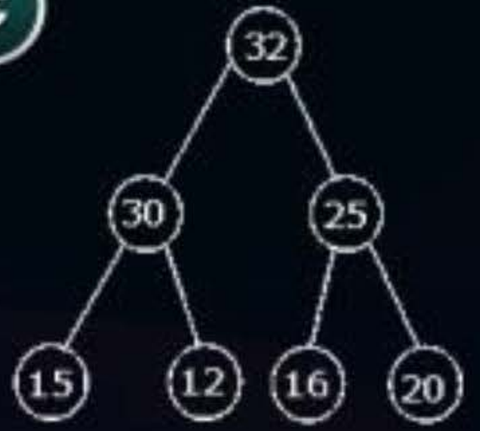
A



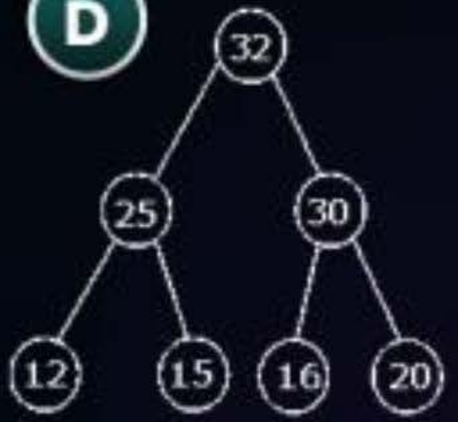
B



C



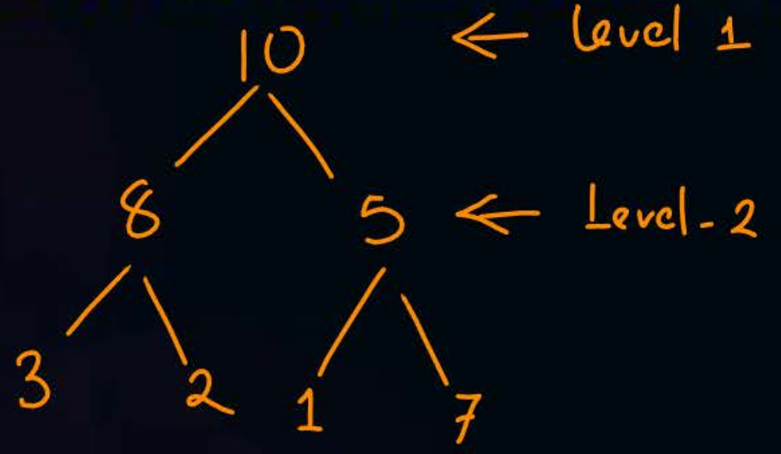
D

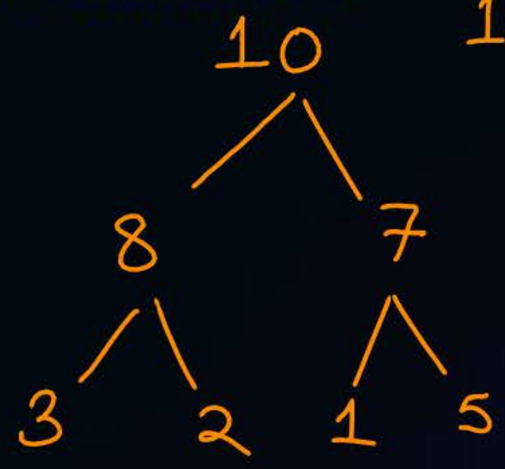




Topic : Heap

#Q. A priority-queue is implemented as a Max-heap. Initially, it has 5 elements. The level-order traversal of the heap is given below:
10, 8, 5, 3, 2
Two new elements '1' and '7' are inserted in the heap in that order. The level-order traversal of the heap after the insertion of the elements is:





10, 8, 7, 3, 2, 1, 5

(A) 10, 8, 7, 5, 3, 2, 1

(B) 10, 8, 7, 2, 3, 1, 5

(C) 10, 8, 7, 1, 2, 3, 5

(D) 10, 8, 7, 3, 2, 1, 5

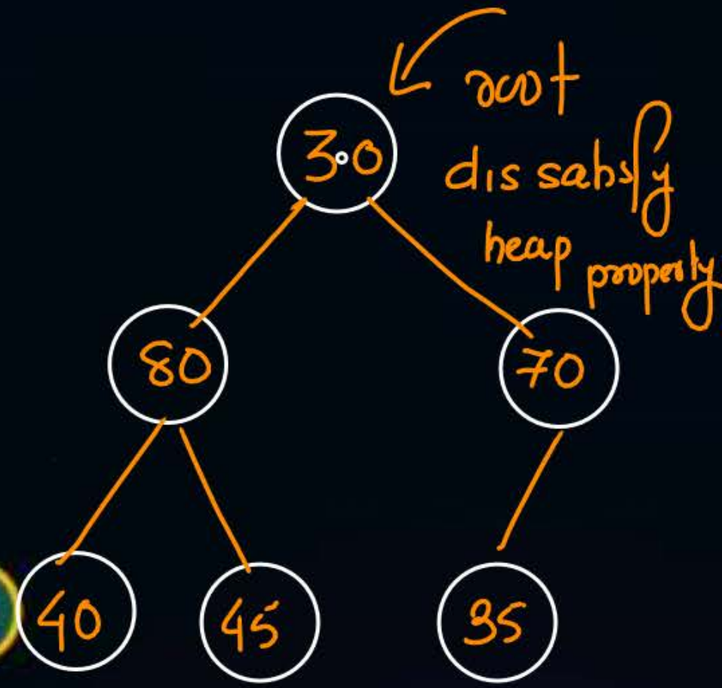


Topic : Heap

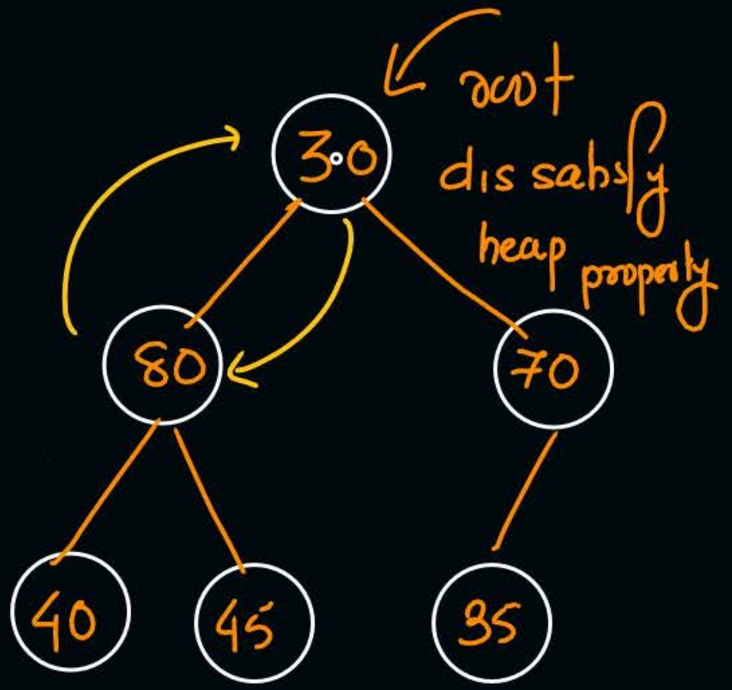
Delete 90



* Delete the Root
* copy last as root
* as we copy the last element as root only root will dissatisfy Heap property



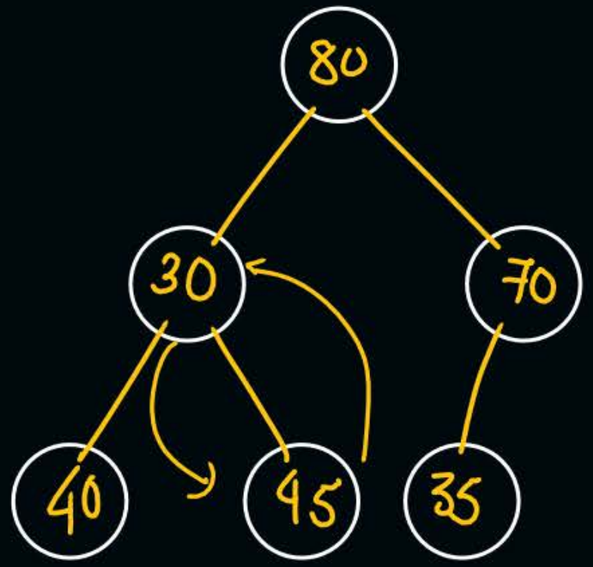
* will Adjust the Root So that it tranform to Max heap again.

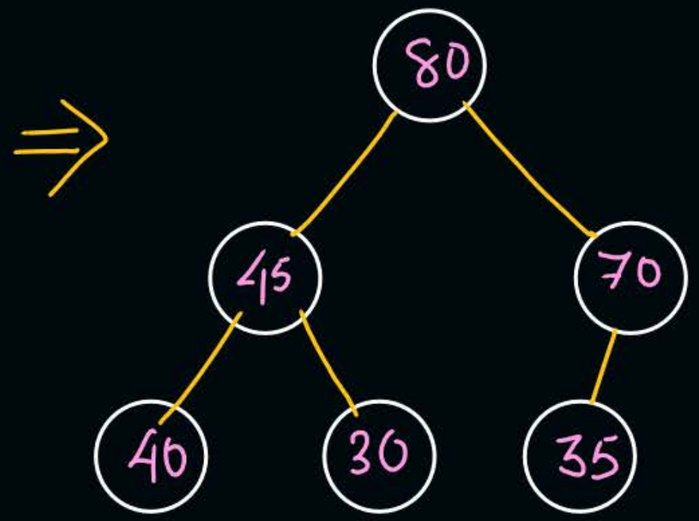


find max of left & right child

then perform the swap if root is lesser for Max heap

repeat the process (comparing left & right child find greater, perform swap)

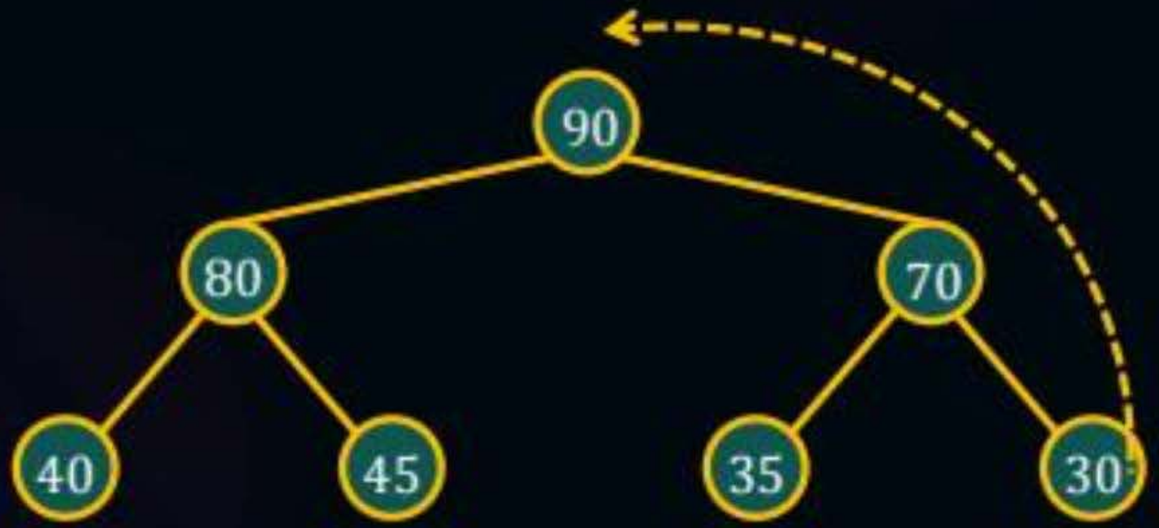






Topic : Heap

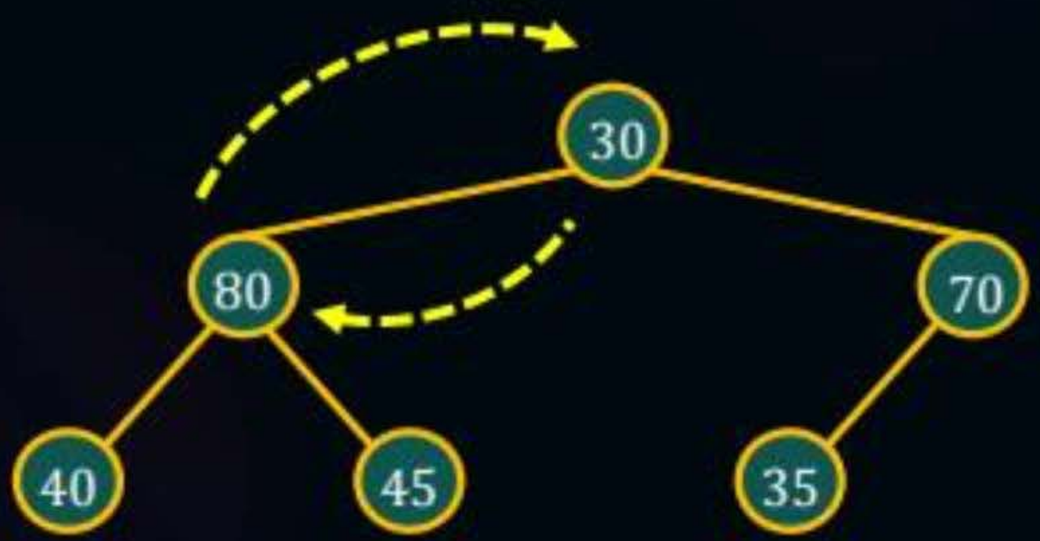
Delete 90





Topic : Heap

Delete 90



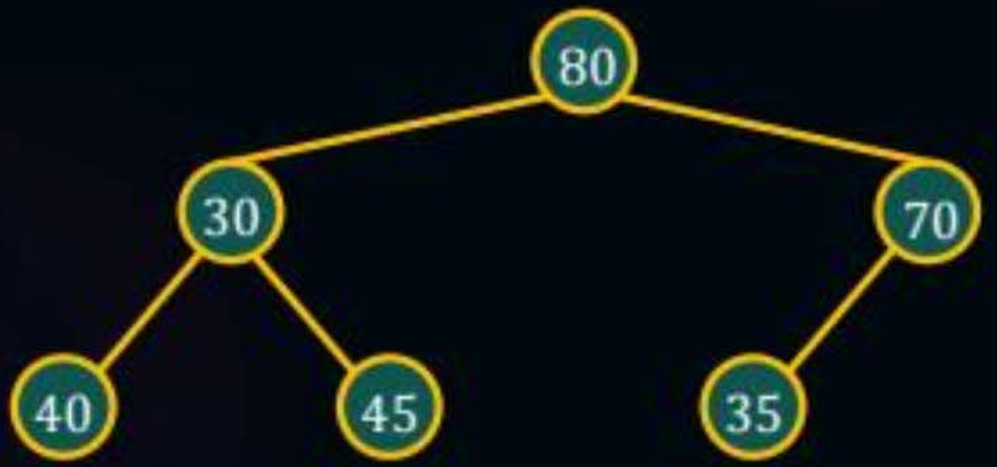
find greater of Left and right children

if root is lesser (Max heap) then swap



Topic : Heap

Delete 90



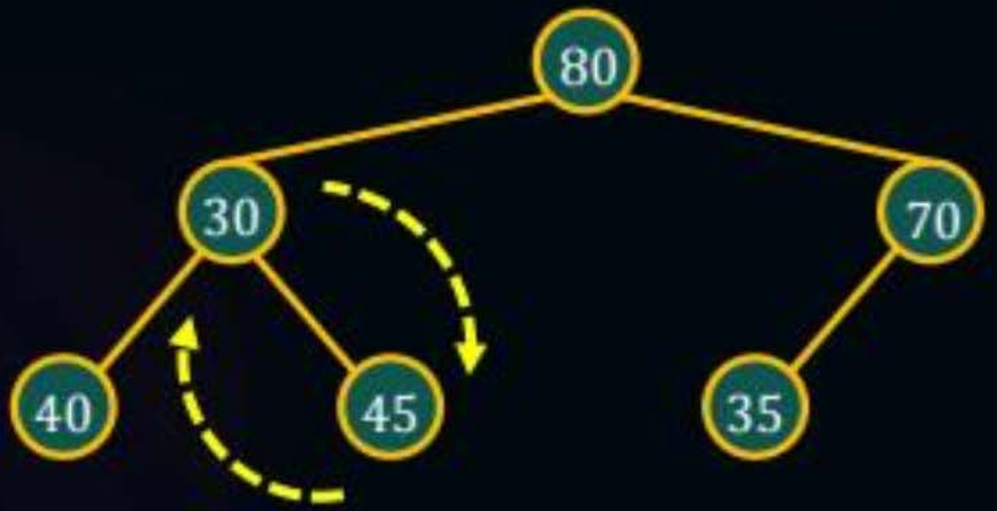
repeat the

same process



Topic : Heap

Delete 90





Topic : Heap

Delete 90



Max swap

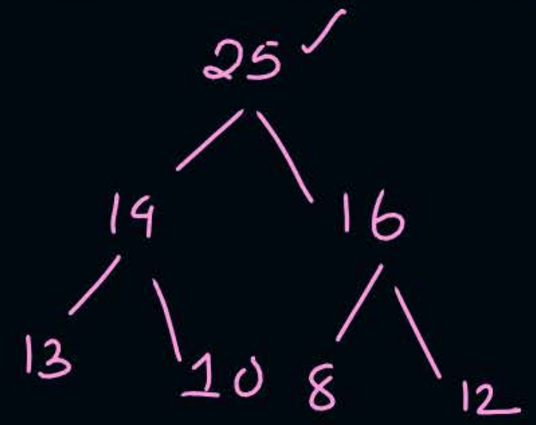
in deletion

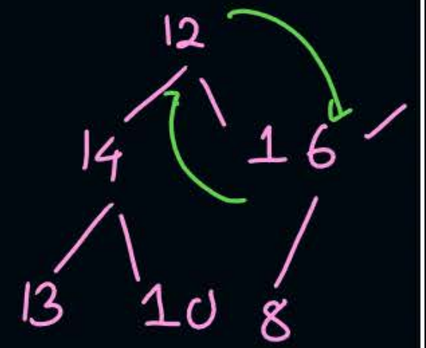
depends upon height

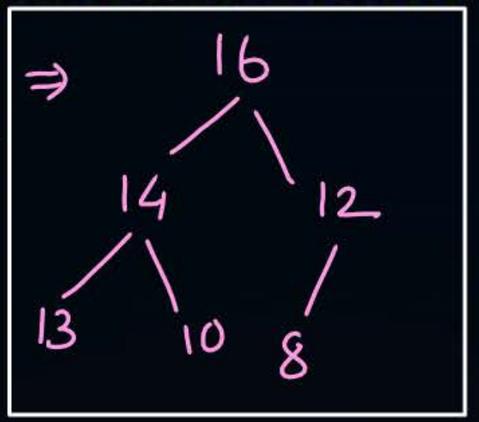
$O(\log_2 n)$

Question

#Q. If heap is represented by {25,14, 16, 13, 10,8 ,12}
What is the content of the array after two delete operations on the correct answer to the previous question:







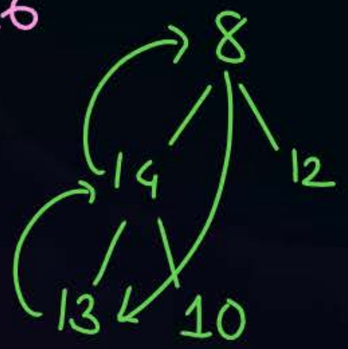
(A) {14, 13, 12, 10, 8}

(B) {14, 12, 13, 8, 10}

(C) {14, 13, 8, 12, 10}

(D) {14, 13, 12, 8, 10}

delete 16





14, 13, 12, 8, 10



2 mins Summary

Topic CBT definition

Topic Height of CBT, No. of Internal Node $\lfloor n/2 \rfloor$

Topic No. of leaves $\lceil n/2 \rceil$

Topic Heap. Max, Min

Topic Insert & delete

Slide

THANK - YOU

Slide